Labwork 4

1. Linear regression

The following table gives the metabolism rate as a function of mass in some animals. Use it to predict the metabolism rate of a 350 – kg tiger.

Animal	**Mass (kg)**	**Metabolism (watts)**
Cow	400	270
Human	70	82
Sheep	45	50
Hen	2	4.8
Rat	0.3	1.45
Dove	0.16	0.97

Do the regression by 2 methods:

(a) Write a script file which employs the following function file:

```
function [ a0,a1,r2 ] = linear_regression( x, y )
%UNTITLED2 Summary of this function goes here
%   Detailed explanation goes here
% Linear regression: y = a0 + a1.x
% x: independent variable
% y: dependent variable
% r2: coefficient of determination
if length(x)~=length(y)
    fprintf('The independent and the dependent variables
should have same number of elements');
else
    n=length(x) % The number of data
end
X=mean(x); Y=mean(y);
a1=(x*y'/n-X*Y)/var(x);
a0=Y-a1*X;
r2= (n*x*y'-n*X*n*Y)/sqrt((n*x*x'-(n*X)^2)*(n*y*y'-(n*Y)^2));
% Create the best fit line
xp=linspace(min(x),max(x),2);
yp=a0+a1*xp;
plot(x,y,'o',xp,yp,'b');
end
```

(b) Use the **Curve fitting Apps** in MATLAB. Is there any difference in the two regression results?

(c) Use the functions `polyfit` in MATLAB to fit the points by a polynominal with degree n and `polyval` to estimate the metabolism rate of the tiger. From what degree of the polynominal the estimated result becomes badly conditioned? And from what degree of the polynominal the fitting curve is no longer unique?

(d) Do the tasks cited in (c) by using the **Curve fitting Apps**.

2. Nonlinear regression

One of the equations approximating the relation between the pressure and the volume of a real gas is as follows:

$$\frac{PV}{RT} = 1 + \frac{A_1}{V} + \frac{A_2}{V^2}$$

The pressure and the volume of the gas are measured and the results are shown in the following table:

P(atm)	0.985	1.108	1.363	1.631
V(mL)	25000	22200	18000	15000

Knowing that R=82.05 mL.atm/gmol.K and T=303K, use the function `fminsearch` to write a script file using the following function file to find A_1 and A_2.

```
function f = PVSSR( a, x, y )
%UNTITLED4 Summary of this function goes here
%   Detailed explanation goes here
if length(x)==length(y)
    n=length(x);
else
    fprintf('The two data must have the same length');
end
yp=zeros(1,n);
for j=1:n
    yp(j)=82.05*303*(1+a(1)/x(j)+a(2)/x(j)^2)/x(j);
end
f=(y-yp)*(y-yp)';
end
```

Explain: what is the meaning of a, x, y and f?

Use the function `fminbnd` to find the minimum value of the pressure if the volume varies between 2000 mL to 4000 mL.

3. Descriptive statistics in MATLAB

a. Statistical functions in MATLAB

Given a set of data s under form of a vector, the functions used to find the characterisitics of this set of data in MATLAB are:

```
mean(s)    median(s)    mode(s)  min(s)    max(s)
var(s)     std(s)
```

MATLAB is also able to generate a histogram from a set of data. The function that is utilized is `hist`. The formula for `hist` is:

```
[n,x]=hist(s)
```

Where n = the number of elements in each bin, x = a vector which elements are the midpoints of each bin, s = the analyzed vector.

The table below shows the height (in cm) of 30 students:

176	183	152	164	159	154
167	168	156	155	153	190
182	178	159	161	166	172
174	175	168	185	184	159
164	162	164	175	170	169

Find

(a) the average height of the 30 students,

(b) the variance

(c) the standard deviation of the distribution

(d) the lowest

(e) the highest heights of the student in this class

(f) the mode and (g) the median height.

Draw a histogram of the height of the students by dividing the students into 5 groups.

b. Uniform distribution and normal distribution in MATLAB

In MATLAB, the functions `rand` and `randn` are used to generate the uniform and normal distribution, respectively.

Using the functions `rand` and `randn`, establish a 100 - elements row vector A whose elements are randomly and uniformly distributed between 140 and 180 and a 200 - elements column vector B whose elements are randomly and normally distributed with mean 170 and standard deviation 30.

Find the average, the variance, the standard deviation, the median and the mode among the elements in A and B.